# Building OpenCV for Windows

Posted by huantran on May 16, 2020 • Edit Post

In this post, I will show how one can build the OpenCV library on a machine running Windows

## Materials

- OpenCV source code
- CMake
- Visual Studio (any version)

## Getting the source code

I will use OpenCV 4.2.0 in this example since it is currently the latest version of OpenCV. Alternatively, one can use the latest version of OpenCV from the master branch of OpenCV on the Github Website. Bootstrapping various the source code is your dream location on the disk. This might not be necessary but choosing a path that has no spaces will be easier for later use of the library.

Download the Cmake for windows

## Getting CMake

CMake is a tool that will help you generate a Visual Studio project for the OpenCV source code. It is a cross platform tool that can also generate Makefiles for Linux and Mac OS and other Build Automation tools. For Windows, I have been comparing the OpenCV library using other tools before the Cygwin, but I after get compared and so this the entire building process listed because there is a space in the path or one of the in the source code, etc.

Back to the tutorial, CMake can be obtained from its official website

## Getting Visual Studio

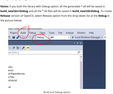
Get this right from Microsoft's Visual Studio page

## Generating Visual Studio project for OpenCV

As mentioned, in this step, we will use CMake to generate a Visual Studio project for the OpenCV source code. After unzipping, what we will get is the extracted folder and we lack the binaries, documentation, and other necessary files that are generated in folder **opencv**. We can see what we obtain from the Build Windows Tab the Name.

Let's create a new folder name **build** and a folder to get generated binary file files.

Run CMake and enter the path in the source folder in the line **Where is the source code** and point the path to **build**, type in the **Where to build the binaries**




Click Add

Click after entering the path

Click **configure** and wait until all the configurations are loaded, the **Generate** in the process. Check and fix configuration, use the configuration in the process below. **Remember** to select the Visual Studio that you installed in your computer in the first selection box. I have Visual Studio 2019 installed on my computer, so I will select Visual Studio 16 2019 and Win32 (VS. select accordingly. Also, if you can be going to compile for a64bit machine, select the x64 in the second box.




Select generating of Visual Studio

After finishing configuring and generating, if you go to the **build** same folder, you will see a project of the selected Visual Studio version


Name
Date modified
Type
2020-05-16 11:00 AM
File folder
2020-05-16 11:00 AM
File folder
2020-05-16 11:00 AM
Visual Studio Solution
2020-05-16 11:00 AM
Visual C++ Project
2020-05-16 11:00 AM
Visual C++ Project
2020-05-16 11:00 AM
Visual C++ Project